The Deterministic Horizon

When Extended Reasoning Fails and Tool Delegation Becomes Necessary

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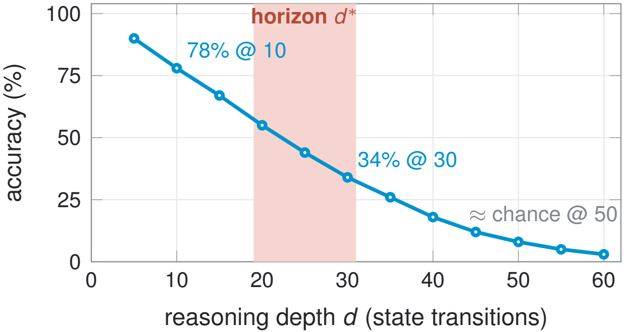
Longer reasoning makes *exact* tasks worse, not better

Past $\sim$20 exact reasoning steps, attention can no longer track state, so **delegate the computation to a tool**.

In **deterministic state tracking** ($\sigma_0 \to \tau$ via exact operators), *one wrong step is failure*, with no partial credit.

Puzzles BFS solves in $<$0.1 s make state-of-the-art models fail after *minutes* of deliberation.

The long reasoning **causes** the failure; it does not cure it.



Two explanations that make *opposite* predictions

PREFERENCE (prior work)

Models *could* track state but *prefer* short reasoning.

⇒ prompting / training should **recover** accuracy.

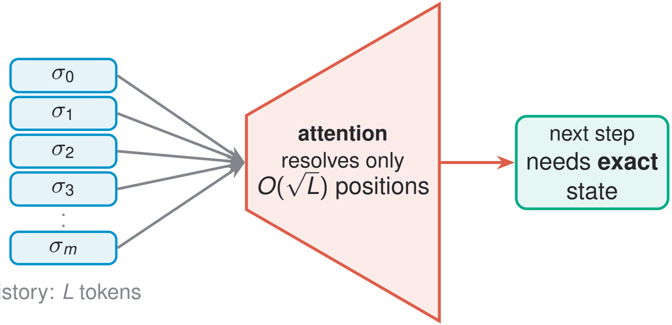
CAPABILITY (this work)

Attention *cannot* hold exact state past a limit: *decoherence*.

⇒ a hard **architectural ceiling** remains.

We lay out **four** contrasting predictions (fine-tuning, length prompts, cross-model correlation, encoder–decoder) and one **direct state-overlap test**. The rest of the talk resolves them.

Why: attention squeezes all history through a narrow channel



Attention Bottleneck Theorem (achievability construction)

$$|\mathcal{S}_{\text{track}}| \le c \cdot 2^{H \log_2(L/H)\sqrt{d_h}} \Rightarrow O(H \log(L/H)\sqrt{d_h})$$

Capacity is huge, but the **per-step** bits for exact tracking cannot fit through the channel. Bound is robust ($<20\%$ shift under $\pm 20\%$ perturbation).

So error grows with depth $\Rightarrow$ a closed-form horizon

Attention entropy rises with position, so per-step error is *not* constant:

$$\epsilon(d) = \epsilon_0 + \gamma \, \frac{d}{L_{\text{eff}}}.$$

Survival probability then decays **super-exponentially**:

$$P(\text{correct at } m) \le \exp\Big(- m\epsilon_0 - \tfrac{\gamma m(m+1)}{2L_{\text{eff}}} \Big).$$

Fine-tuning ceiling: for $d > d^*$, $\text{Acc}_{\text{ft}} \le \text{Acc}_{\text{base}} + O(d^*/d)$; no training distribution escapes it.

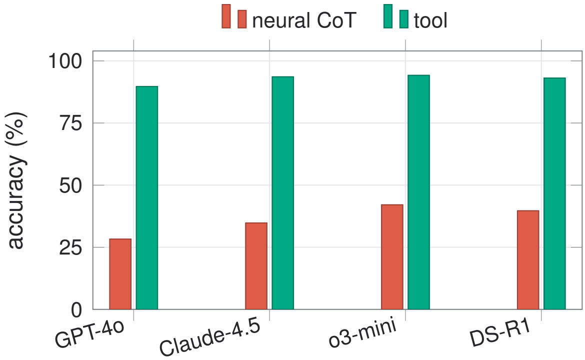
DETERMINISTIC HORIZON

$$d^* = \frac{-\epsilon_0 L_{\text{eff}} + \sqrt{\epsilon_0^2 L_{\text{eff}}^2 + 2\gamma L_{\text{eff}} \ln(1/\alpha)}}{\gamma}$$

primary suite $\boxed{d^* \in [19, 31]}$

scales as $d^* \propto \sqrt{d_h H}$ (7–8B: $\sim$19–20 | 70–72B: $\sim$28)

Evidence 1: tool delegation 86–94% vs neural 24–42%



720,000

evaluations: 12 models $\times$ 5 conditions $\times$ 8 tasks $\times$ 500 instances $\times$ 3 seeds (real tasks incl. SWE-Bench, WebArena, SQL-Multi)

Cohen's $d = 2.1$–3.4 (very large); consistent across all model families.

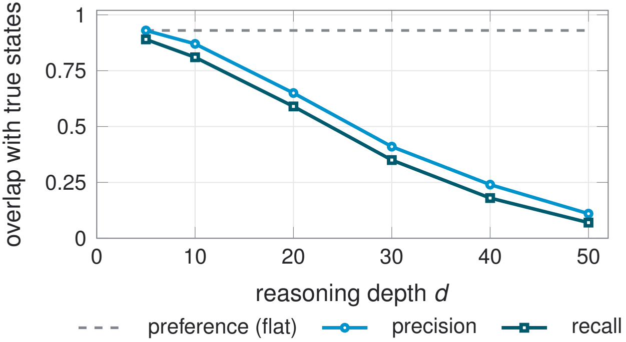
4.2–4.7$\times$ cheaper per correct answer.

Evidence 2: it is architectural, and all four predictions confirm

Discriminating test	Preference	Capability (ours)	Observed
Fine-tuning recovery	$>30\%$	$<5\%$	**3.2%** ✓
"Reason longer" prompt	$>10\%$	$<2\%$	**0.9%** ✓
Cross-model correlation	low	high	$r = 0.81$–0.91 ✓
Encoder–decoder edge	none	2–3$\times$	**2.8**$\times$ ✓

Preference levers (prompting, RLHF, fine-tuning)
cannot lift an architectural ceiling.

Direct test: the model drifts into states that don't exist

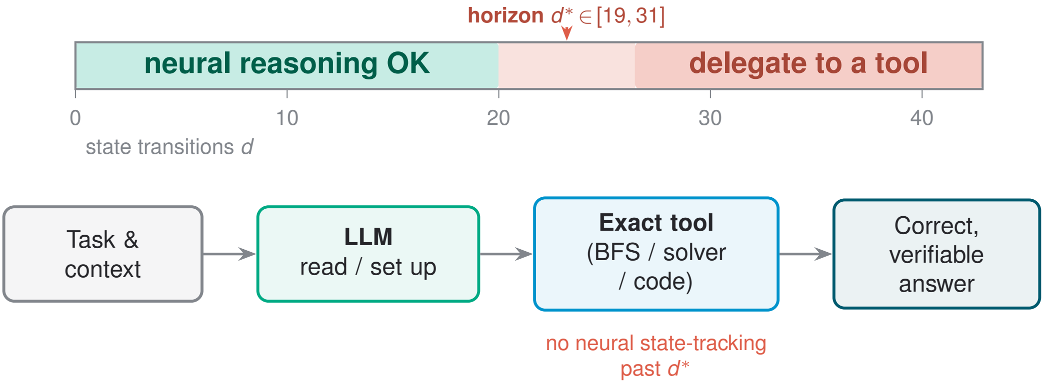


State-Space Jaccard compares the truly reachable states with the states the model claims.

A mere *preference* would keep **precision flat** (dashed).

Instead **both** precision *and* recall collapse together ($R^2 = 0.99$ super-exponential fit): the signature of **capability** failure, not truncation.

What to do: cross the horizon ⇒ delegate



Delegate: program-state tracing, long puzzles, multi-hop tracking, many-table SQL joins.

Keep neural: short or approximate tasks (arithmetic, retrieval, summarization).

Takeaway

Decoder-only attention has a state-tracking ceiling

Explicit state registers & adaptive tool routing

Delegate by default past $\sim$20 exact transitions

What would change the picture: an architecture that holds exact state past the horizon, or tool-free reasoning that matches **86–94%** at depth $> d^*$. Until then, beyond **$\sim$20 transitions** the reliable choice is to delegate.

Thank you.